AmiDrive

OneDrive client for AmigaOS 3.x – how it works, how it runs on the A1200 and a guide for a new user

Program version 0.4.2, document of 18 October 2026.

1. What AmiDrive is

AmiDrive is a native Amiga program that synchronises **one chosen OneDrive folder** with a directory on the Amiga's disk. It connects directly to Microsoft Graph over the Internet (HTTPS, TLS 1.2/1.3) – no PC, Raspberry Pi or rclone needed. Files put into that folder on a phone or PC appear on the Amiga, and files changed on the Amiga go to the cloud.

- Amiga: AmigaOS 3.x, 68020 or faster, aims at 2–4 MB of Fast RAM.
- Network: any TCP/IP stack with `bsdsocket.library` (Roadshow, AmiTCP, Miami); encryption by AmiSSL 5.
- Sign-in with a device code: the Microsoft account password never reaches the Amiga.
- Runs in the background: checks for changes every few minutes, only the differences since the last time.

2. How it works

One **synchronisation pass** goes like this:

1. **Token.** The program reads the token from `ENVARC:AmiDrive/token`; when it has expired (after about 1 h) it renews it at Microsoft and saves the new one.
2. **Changes in the cloud.** A `/delta` request for the synchronised folder. The first time OneDrive returns the whole content, later only the differences since the remembered point (deltaLink). Answers are read in pages of 20 items to fit in little memory.
3. **Download to the Amiga.** New and changed files are streamed into a temporary file in the same directory and only then put in place; the file date is the one from the cloud. Moves and renames are done locally, without downloading again. Deleted in the cloud – deleted on the Amiga.
4. **Changes on the Amiga.** The program scans the local directory and compares it with the state database (size and date of each file after the last synchronisation). New files and folders are uploaded, changed files are uploaded as new versions, moved ones are moved in the cloud, deleted ones are deleted (they go to the OneDrive recycle bin).
5. **State.** The state database (`.AmiDrive/state`) is saved safely: into a temporary file, then swapped in. The new delta point is remembered only after a pass without errors – a file that failed comes back in the next pass.

Conflicts and data safety

Situation	What AmiDrive does
File changed both on the Amiga and in the cloud	The cloud version gets the proper name, the Amiga version stays as *name-Amiga.ext* and is uploaded as a separate file. Nothing is lost.
File deleted on the Amiga but changed in the cloud	The file comes back from the cloud.
File deleted in the cloud but changed on the Amiga	The file on the Amiga stays.
Most files missing locally (e.g. a disk not mounted)	When more than 10 items and more than half are missing, nothing is deleted in the cloud and the log has a warning.
Interruption (Ctrl-C, reset, no network)	The state stays consistent; the next pass finishes the interrupted work.

File names

- OneDrive uses UTF-8, the Amiga Latin-1 or Latin-2 (`CHARSET=LATIN2` for Central European letters). Characters outside the character set become "_".
- Names longer than the disk allows (OFS/FFS: 30 characters) are shortened to *start~hhhhhh.ext*; the full cloud name is kept in the state database.
- Amiga protection bits and file comments are not transferred (OneDrive does not know them).
- Skipped: OneNote notebooks, items shared from someone else's drive, files of 4 GB and more, the `.AmiDrive` directory.

3. How AmiDrive runs on the author's A1200

State of 18 October 2026 (A1200, 68060, AmigaOS 3.2.2, Roadshow, AmiSSL 5.24, OneDrive for Business):

What	Where / value
Program and icon	`Work:AmiDrive/AmiDrive`, `Work:AmiDrive/AmiDrive.info`
Autostart	`SystemSD:WBStartup/AmiDrive.info` (project icon: default tool `Work:AmiDrive/AmiDrive`, DONOTWAIT) and a short text file `SystemSD:WBStartup/AmiDrive`
Synchronised cloud folder	`/Retro/AmigaDrive`
Directory on the Amiga	`Work:OneDrive`
Settings	`ENVARC:AmiDrive/AmiDrive.prefs`: LOCAL, REMOTE, CHARSET=LATIN2, INTERVAL=3, PAUSE=YES (upload on – the default)
Sign-in token (secret)	`ENVARC:AmiDrive/token`
State and log	`Work:OneDrive/.AmiDrive/state`, `Work:OneDrive/.AmiDrive/AmiDrive.log`
Time zone	`TZ=CET-1CEST,M3.5.0,M10.5.0/3` (with summer time)

How it is started

AmiDrive **starts by itself with the Workbench**: `SystemSD:WBStartup` holds a project icon that starts the program in `Work:AmiDrive` (no second copy of the program – a new version in `Work:AmiDrive` is used by the autostart at once). The network is already up then (`S:User-Startup`). The Tool Types of the program icon are switched off (in brackets), so the settings come from the prefs file. The program runs **in the background, without a window**. `PAUSE=YES` is set: after the start AmiDrive **waits paused** ("Paused", as after *Stop*) and checks and sends nothing until *Sync now* in the window or the *Tools* menu; from then on it makes a pass every 3 minutes and writes it to the log. A window appears only for an error or a sign-in.

The "OneDrive" icon on the Workbench

While AmiDrive runs in the background, the Workbench shows a **OneDrive** icon (a cloud; with orange round arrows during a synchronisation):

- **double-click** opens the synchronised directory `Work:OneDrive` (AmigaOS 3.5 or newer; from 3.9/3.2 as a list of all files by name) and the **AmiDrive window** with the buttons *Sync now*, *Stop* (pauses the synchronisation) and *Quit* (ends the program) – at once, also during a synchronisation; then the window shows the progress: a bar with percent, the current file, the speed and the estimated time left (Workbench 3.2 has no icon menu on the right button);
- **files and drawers dropped on the icon** are copied into that directory and uploaded to the cloud at once;
- *Tools* menu: **AmiDrive: status** (the progress window, or the result of the last check and the time of the next one), **AmiDrive: synchronise now** (a pass now) and **AmiDrive: quit** (ends the background work, the icon goes away).

The Workbench chooses the place of the icon (the first free one in the icon column – it may be hidden by a window). Without the icon: `APPICON=NO`.

Everyday use

Task	How
Run in the background	Double-click the *AmiDrive* icon, or in a Shell: `Run >NIL: Work:AmiDrive/AmiDrive`
End it	The *Quit* button, menu *Tools* → *AmiDrive: quit*, double-click the program icon again → *Stop*, or in a Shell `Work:AmiDrive/AmiDrive STOP`
One pass now	Menu *Tools* → *AmiDrive: synchronise now* (while it runs in the background), or `Work:AmiDrive/AmiDrive ONCE`
What happened	`Type Work:OneDrive/.AmiDrive/AmiDrive.log`
Account and settings	`Work:AmiDrive/AmiDrive INFO`
Switch off the autostart	Delete `SystemSD:WBStartup/AmiDrive` and `AmiDrive.info` from that directory

Only one synchronisation can run at a time: a second copy started from a Shell ends with “already running”, one started from the icon asks whether to stop the running one.

4. What another user has to do (their own OneDrive)

Requirements

- AmigaOS 3.x (tested on 3.2.2), 68020 or newer, a few MB of Fast RAM.
- A working TCP/IP stack with `bsdsocket.library` v4 (Roadshow recommended) and an Internet connection (network card, PCMCIA WiFi, Plipbox…).
- AmiSSL 5.24 or newer (with the certificates in `AmiSSL:Certs`).
- A Microsoft account with OneDrive: personal (outlook.com, hotmail, live) or work/school (Microsoft 365).

Step by step

1. **Clock and time zone.** Set the correct date and time (without it TLS rejects the servers' certificates). For example Central Europe with summer time:

```
SetEnv SAVE TZ CET-1CEST,M3.5.0,M10.5.0/3
```

2. **Installation.** Copy `AmiDrive` and `AmiDrive.info` to any directory, e.g. `Work:AmiDrive`.
3. **Folder in OneDrive.** In the browser or the OneDrive app create the folder to be synchronised, e.g. `Amiga`. It must exist – AmiDrive does not create it (message “OneDrive folder … not found”). Best start with a folder with few files.
4. **Sign-in.** In a Shell `AmiDrive LOGIN` (or just click the icon – without a sign-in the program opens a window by itself). On a phone or PC open `https://login.microsoft.com/device`, enter the code from the Amiga screen, sign in and allow the *AmiDrive* app to access your files. The token is saved in `ENVARC:AmiDrive/token`.
5. **Settings** (remembered):

```
AmiDrive LOCAL=Work:OneDrive REMOTE=/Amiga CHARSET=LATIN2 ONCE
```

This first pass downloads the content of the folder. Instead of the Shell the same options can be put in the Tool Types of the icon (*Icons* → *Information…*, remove the brackets).

6. **Background and autostart.** Click the icon or `Run >NIL: AmiDrive`. Autostart: copy the program with its icon to `SYS:WBStartup` (the icon has DONOTWAIT) or – as on the A1200 – put a project icon there with the default tool pointing at the program (`tools/make_icon.py --project`). To make AmiDrive wait for *Sync now* after the start: `AmiDrive PAUSE=YES`.

The installer (*Install-AmiDrive*) asks for all of this by itself: the program drawer, the directory for the files, the OneDrive folder, the character set, how often to check, the direction, the “OneDrive” icon, the clock, the time

zone, the autostart and what to do after the start (synchronise at once or wait paused until *Sync now*).

Several OneDrive accounts

AmiDrive handles up to 3 accounts (e.g. work and personal); one is synchronised at a time. In the active AmiDrive window the right mouse button shows the **Account** menu: the accounts (the active one ticked, “(start)” at the start account), *Start with this account* (AmiDrive always starts with it; without it – with the last one used), *Sign in...* and *Sign out*. Choosing an empty account shows the sign-in code in the window, then asks for the drawer on the Amiga and the folder in that OneDrive. Each account has its own drawer, folder and token (account 1: `AmiDrive.prefs` and `token`, accounts 2–3: `Account2.prefs`, `token2` …). In a Shell: `AmiDrive ACCOUNT=2`.

Personal and work accounts – the app registration

AmiDrive signs in through an app registered in Microsoft Entra ID (the ID built into the program: `0474e444-3266-446b-b49c-d47a01cadf56`, for personal accounts and work accounts of any organisation).

- **Personal account** – just sign in and accept the consent.
- **Work account of another organisation** – the administrator may block apps from outside the organisation. Then the sign-in ends with a request for admin consent. Solutions: the administrator approves the *AmiDrive* app (Enterprise applications → Grant admin consent), or the organisation registers its own app as below.

Your own app registration (Entra portal or Azure Portal):

1. *App registrations* → *New registration*; name e.g. AmiDrive; account types: this organisation only, or any organisation and personal accounts.
2. *Authentication* → *Advanced settings* → *Allow public client flows: Yes* (device code sign-in, no secret).
3. *API permissions* → *Microsoft Graph* → *Delegated*: **Files.ReadWrite** and **offline_access** (optionally admin consent for the whole organisation).
4. Copy the *Application (client) ID* and the *Directory (tenant) ID*, then on the Amiga:

```
AmiDrive CLIENTID=<client-id> TENANT=<tenant-id or organizations>
AmiDrive LOGIN
```

Speed

On the Amiga the speed is limited by the encryption, not by the network. Measured on the A1200 (68060/50, Ethernet of the Warp card), the same 5 MB file:

Connection	Speed
no encryption (http)	1590 KB/s
TLS 1.3, AES-256-GCM	215 KB/s
TLS 1.3, ChaCha20-Poly1305	337 KB/s
OneDrive, full synchronisation of 20 MB (AES-128 – Microsoft's choice)	192 KB/s (108 s; with AES-256: 130 s)

AmiDrive offers the ciphers fastest on the 68k (ChaCha20, then AES-128), all fully secure. Your own measurement: `AmiDrive TEST URL=https://…`.

Security

- The file `ENVARC:AmiDrive/token` gives access to the files in OneDrive – do not share it or copy it to other computers. `AmiDrive LOGOUT` deletes it from the Amiga; access can also be revoked in the Microsoft account (Apps and services).
- Server certificates are always checked; tokens never go into the log.

5. Options

The same in a Shell and in the Tool Types of the icon. Precedence: Shell, then Tool Types, then the settings file. Only options given in a Shell are saved.

Option	Meaning
LOCAL=directory	directory on the Amiga, e.g. Work:OneDrive (created when missing)
REMOTE=/folder	folder in OneDrive, e.g. /Amiga (must exist)
CHARSET=LATIN1|LATIN2	character set of names on the Amiga (default LATIN1)
INTERVAL=min	minutes between checks in the background (default 3)
UPLOAD=YES|NO	NO – download from the cloud only, no upload (default YES)
STATE=directory	another place for the state database and the log (default LOCAL/.AmiDrive)
ONCE	one pass, then end
WINDOW	(from the icon) a window with the progress
APPICON=YES|NO	the “OneDrive” icon on the Workbench while running in the background (default YES)
PAUSE=YES|NO	YES – start paused (“Paused”), the first check only after *Sync now* (default NO); the installer asks about it too
CLOCK=ASK|AUTO|NO	wrong Amiga clock: a window asking (default) / set it at the start and once a day / do not check
SETCLOCK	sets the Amiga clock (and the battery clock) from the Internet
STOP	stops the AmiDrive running in the background
LOGIN / LOGOUT / INFO	sign in, sign out, account and settings
TEST	checks the clock, time zone, DNS, TLS and the connection to Microsoft Graph
CLIENTID= / TENANT=	your own app registration (chapter 4)
ACCOUNT=1|2|3	the active OneDrive account (remembered); in the window: menu Account
START=0|1|2|3	the account to start with (0: the last one used); menu Start with this account

6. Problems and limits

Message / symptom	Cause and advice
Not signed in	no token: AmiDrive LOGIN
Sign in again	the token expired or access was revoked (e.g. password change): AmiDrive LOGIN
OneDrive folder … not found	wrong REMOTE path or the folder does not exist
certificate rejected, check the Amiga clock	wrong Amiga clock: AmiDrive shows a window with the time from the Internet and a *Set clock* button (CLOCK=ASK) or sets it by itself (CLOCK=AUTO); by hand: AmiDrive SETCLOCK
certificate rejected (other)	missing or old AmiSSL certificates – update AmiSSL; AmiDrive TEST
already running	another copy is synchronising: AmiDrive STOP
A file ending in -Amiga	a conflict – both versions are kept, choose the right one

- An interrupted download or upload of a large file starts again from the beginning (no resume).
- Renaming a folder on the Amiga: a new folder is created in the cloud, the files are moved into it (without uploading), the old one is deleted.
- One synchronised folder; files up to 4 GB; summer time only from the TZ variable.
- An AmiDrive process hung after a crash blocks a new start until the Amiga is restarted.